RedSim - Security Analysis & Education Platform

RedSim is a comprehensive web-based platform built with Laravel that provides various cybersecurity tools, AI-powered security analysis, and interactive educational features.

🚀 Features

🛡️ Security Analysis

- **Analisis Kode (Code Analysis):** Inspect source code for vulnerabilities.
- **Analisis URL (URL Analysis):** Verify and scan URLs for malicious content. Includes domain verification.
- **Analisis ZIP (ZIP Analysis):** Scan archived files for security threats.
- **Analisis Log (Log Analysis):** Parse and analyze log files for anomalies.
- **AI-Powered Analysis:** Deep integration with AI for advanced threat detection and analysis.
- **PDF Reports:** Generate and export comprehensive security scan reports.

🎓 Education & Training

- **Tantangan (Challenges):** Interactive cybersecurity quizzes and challenges.
- **Ensiklopedia (Encyclopedia):** Knowledge base for cybersecurity concepts.
- **Leaderboard:** Gamified learning experience to track progress and rank users.

🔐 Authentication & Security

- **WhatsApp OTP:** Secure login and password reset via WhatsApp OTP.
- **Google OAuth:** Seamless single sign-on (SSO) integration.
- **User Approval System:** Manual verification process by admin for new users.

📊 Admin Dashboard

- **System Health Monitoring:** Keep track of the platform's status.
- **Audit Logs:** Monitor user activities and system events.
- **AI Configuration:** Manage API keys and LLM settings dynamically.
- **User & Content Management:** Manage users, challenges, and approvals.

🤖 AI Chat Assistant

- Integrated AI Chatbot to assist users with cybersecurity queries.
- Retrieval-Augmented Generation (RAG) debugging for administrators.

🛠️ Tech Stack

- **Backend:** Laravel (PHP 8.3+)
- **Database:** MySQL / PostgreSQL
- **Authentication:** Laravel Socialite, Custom WhatsApp Integration

⚙️ Installation

1. **Clone the repository:**

```
git clone https://github.com/barbarbarox/redsim.git
cd redsim
```

2. **Install dependencies:**

```
composer install
npm install
```

3. **Configure Environment:**

```
cp .env.example .env
php artisan key:generate
```

Make sure to configure your database connection and API keys (Google OAuth, WhatsApp API, AI services) in the `.env` file.

4. **Run Migrations & Seeders:**

```
php artisan migrate --seed
```

5. **Serve the Application:**

```
php artisan serve
npm run dev
php artisan queue:work
```

Catatan: Jalankan `php artisan queue:work` di terminal terpisah agar fitur AI dan proses latar belakang lainnya dapat berjalan dengan lancar dan aman.

📝 License

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